

Statistics & Probability



Mean, median, and mode

- 1 Find the mean of 4, 8, 6, 10, 7.
 - 2 Find the median of 3, 9, 5, 1, 7.
 - 3 Find the mode of 2, 4, 4, 6, 8, 4, 2.
 - 4 A data set has values 8, 12, 10, 15, 10, 14, 11. Find the mean, median, and mode.
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Range and quartiles

- 5 Find the range of 5, 12, 8, 20, 3.
 - 6 For the data set 2, 4, 5, 7, 8, 9, 12, find Q_1 and Q_3 .
 - 7 Find the interquartile range (IQR) for the same data set.
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Variance and standard deviation

- 8 Find the variance of 2, 4, 6, 8, 10, using $\sigma^2 = \frac{\sum (x - \mu)^2}{n}$.
 - 9 Find the standard deviation of the same data set, rounded to two decimal places.
 - 10 A data set has variance 25. State its standard deviation.
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Frequency tables

- 11 A frequency table shows value 1 with frequency 3, value 2 with frequency 5, and value 3 with frequency 2. Find the mean.
 - 12 Using the same frequency table, find the mode.
 - 13 Using the same frequency table, find the median.
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Basic probability

- 14 A fair six-sided die is rolled. Find $P(\text{number} > 4)$.
- 15 A bag contains 5 red and 3 blue marbles. Find $P(\text{blue})$.
- 16 A card is drawn from a standard 52-card deck. Find $P(\text{king})$.
- 17 Two fair coins are flipped. Find $P(\text{both heads})$.
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Complementary and mutually exclusive events

- 18 If $P(A) = 0.35$, find $P(\text{not } A)$.
- 19 A die is rolled. Events $A = \{\text{even}\}$ and $B = \{\text{rolling a } 5\}$. Are A and B mutually exclusive?
- 20 If $P(A) = 0.4$ and $P(B) = 0.25$, and A, B are mutually exclusive, find $P(A \text{ or } B)$.
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Independent events

- 21 Two independent events have $P(A) = 0.6$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.
- 22 A fair coin is flipped twice. Find $P(\text{first heads and second tails})$.
- 23 A card is drawn from a deck, replaced, then a second card is drawn. Find $P(\text{both are aces})$.
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Conditional probability

- 24 In a class, $P(\text{math}) = 0.7$ and $P(\text{physics and math}) = 0.4$. Find $P(\text{physics} \mid \text{math})$.
- 25 A box has 10 balls: 6 red and 4 blue. Two balls are drawn without replacement. Find $P(2\text{nd is blue} \mid 1\text{st is blue})$.
- 26 Given $P(A) = 0.5$, $P(B) = 0.3$, $P(A \text{ and } B) = 0.15$, find $P(A \mid B)$.
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Counting: permutations and combinations

- 27 In how many ways can 4 books be arranged on a shelf?

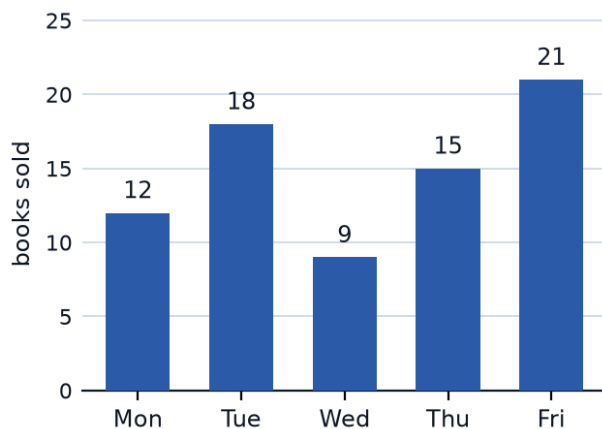
- 28 How many ways can a committee of 3 be chosen from 8 people (order doesn't matter)?
- 29 A code uses 3 distinct digits from 0–9, where order matters. How many possible codes are there?
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The five-number summary

- 30 For the data set 6, 10, 14, 18, 22, 26, 30, find the five-number summary (minimum, Q_1 , median, Q_3 , maximum).
- 31 Find the IQR for the same data set.
- 32 Outlier boundaries are $Q_1 - 1.5 \times \text{IQR}$ and $Q_3 + 1.5 \times \text{IQR}$. Find both boundaries for the same data set.
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Reading data from a bar chart

- 33 The bar chart shows the number of books sold each day at a bookstore over five days.



- a Find the total number of books sold over the five days.
- b Find the mean number of books sold per day.
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Word problems: statistics and probability in context

- 34 This question has two parts. A survey of 200 students found that 120 like math and 90 like science, while 50 like both subjects.
- a Find the probability that a randomly selected student likes math or science, using $P(M \cup S) = P(M) + P(S) - P(M \cap S)$.
- b Find the probability that a student likes neither subject.

- 35 This question has two parts. A quality-control inspector finds that 2% of items from a production line are defective. A random sample of 3 items is tested, and defects occur independently.
- a Find the probability that all three items are defective.
 - b Find the probability that none of the three items are defective.
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MCQ — Mean, median, and mode

- 36 Find the mean of 5, 9, 7, 11, 8.
- a. 8
 - b. 7
 - c. 40
 - d. 9
- 37 Find the median of 4, 10, 6, 2, 8.
- a. 5
 - b. 8
 - c. 6
 - d. 4
- 38 Find the mode of 3, 5, 5, 7, 9, 5, 3.
- a. 5
 - b. 4
 - c. 3
 - d. 7

- 39 A data set has values 9, 13, 11, 16, 11, 15, 12. Find the mean, median, and mode.
- a. Mean = 13, median = 12, mode = 11
 - b. Mean = 12.43, median = 11, mode = 12
 - c. Mean $\approx$ 12.43, median = 12, mode = 11
 - d. Mean $\approx$ 12.43, median = 13, mode = 11
- 40 Find the mean of 12, 18, 15, 21, 14.
- a. 15
 - b. 16
 - c. 80
 - d. 18
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MCQ — Range and quartiles

- 41 Find the range of 6, 14, 9, 22, 4.
- a. 22
 - b. 16
 - c. 18
 - d. 20
- 42 For the data set 3, 5, 6, 8, 9, 10, 13, find Q_1 and Q_3 .
- a. $Q_1 = 5$, $Q_3 = 9$
 - b. $Q_1 = 5$, $Q_3 = 10$
 - c. $Q_1 = 3$, $Q_3 = 13$
 - d. $Q_1 = 6$, $Q_3 = 9$

43 Find the interquartile range (IQR) for the same data set.

- a. 10
- b. 8
- c. 3
- d. 5

44 Find the range of 18, 25, 12, 30, 9.

- a. 18
- b. 16
- c. 21
- d. 30

MCQ — Variance and standard deviation

45 Find the variance of 3, 5, 7, 9, 11, using $\sigma^2 = \frac{\sum (x - \mu)^2}{n}$.

- a. 4
- b. 2.83
- c. 8
- d. 40

46 Find the standard deviation of the same data set, rounded to two decimal places.

- a. 1.41
- b. 4.00
- c. 2.83
- d. 8.00

47 A data set has variance 36. State its standard deviation.

- a. 6
- b. 12
- c. 18
- d. 1,296

48 Find the variance of 4, 8, 12, 16, 20, using $\sigma^2 = \frac{\sum(x - \mu)^2}{n}$.

- a. 160
- b. 5.66
- c. 16
- d. 32

MCQ — Frequency tables

49 A frequency table shows value 1 with frequency 4, value 2 with frequency 6, and value 3 with frequency 5. Find the total number of data points.

- a. 10
- b. 15
- c. 3
- d. 6

50 Using the same frequency table, find the mode.

- a. 6
- b. 3
- c. 2
- d. 1

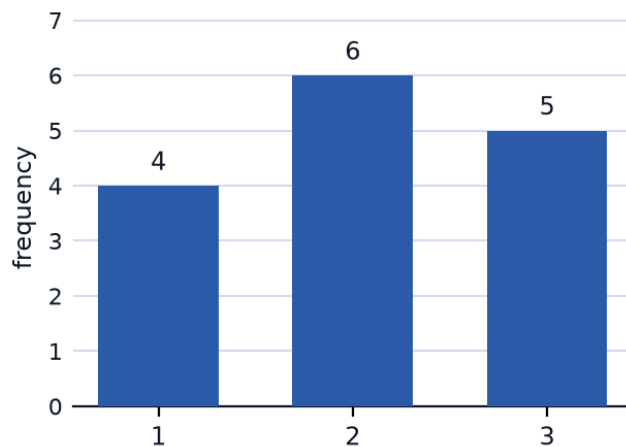
51 Using the same frequency table, find the median.

- a. 1.5
- b. 1
- c. 2
- d. 3

52 Using the same frequency table, find the mean.

- a. 2.33
- b. ≈ 2.07
- c. 2.00
- d. 1.87

53 The bar chart shows a frequency table. Find the total number of data points.



- a. 15
- b. 6
- c. 3
- d. 10

MCQ — Basic probability

54 A fair six-sided die is rolled. Find $P(\text{number} < 3)$.

a. $\frac{1}{6}$

b. $\frac{1}{2}$

c. $\frac{1}{3}$

d. $\frac{2}{3}$

55 A bag contains 7 red and 4 blue marbles. Find $P(\text{red})$.

a. $\frac{7}{11}$

b. $\frac{1}{11}$

c. $\frac{4}{11}$

d. $\frac{7}{4}$

56 A card is drawn from a standard 52-card deck. Find $P(\text{queen})$.

a. $\frac{4}{13}$

b. $\frac{1}{4}$

c. $\frac{1}{13}$

d. $\frac{1}{52}$

57 Two fair coins are flipped. Find $P(\text{at least one tail})$.

a. $\frac{1}{2}$

b. $\frac{3}{4}$

c. $\frac{1}{4}$

d. $\frac{1}{3}$

58 A fair six-sided die is rolled. Find $P(\text{multiple of } 3)$.

- a. $\frac{2}{5}$
- b. $\frac{1}{3}$
- c. $\frac{1}{2}$
- d. $\frac{1}{6}$

MCQ — Complementary and mutually exclusive events

59 If $P(A) = 0.28$, find $P(\text{not } A)$.

- a. 0.28
- b. 1.28
- c. 0.5
- d. 0.72

60 A die is rolled. Events $A = \{\text{odd}\}$ and $B = \{\text{rolling a } 4\}$. Are A and B mutually exclusive?

- a. Yes
- b. Only sometimes
- c. No
- d. Cannot be determined

61 If $P(A) = 0.3$ and $P(B) = 0.45$, and A, B are mutually exclusive, find $P(A \text{ or } B)$.

- a. 0.75
- b. 0.15
- c. 1.05
- d. 0.135

62 If $P(A) = 0.6$, find $P(\text{not } A)$.

- a. 1.6
- b. 0.4
- c. 0.5
- d. 0.6

MCQ — Independent events

63 Two independent events have $P(A) = 0.7$ and $P(B) = 0.4$. Find $P(A \text{ and } B)$.

- a. 1.1
- b. 0.28
- c. 0.3
- d. 0.175

64 A fair coin is flipped twice. Find $P(\text{both heads})$.

- a. $\frac{1}{3}$
- b. 1
- c. $\frac{1}{4}$
- d. $\frac{1}{2}$

65 A card is drawn from a deck, replaced, then a second card is drawn. Find $P(\text{both are kings})$.

- a. $\frac{2}{13}$
- b. $\frac{1}{26}$
- c. $\frac{1}{13}$
- d. $\frac{1}{169}$

- 66 Two independent events have $P(A) = 0.5$ and $P(B) = 0.8$. Find $P(A \text{ and } B)$.
- a. 0.65
 - b. 1.3
 - c. 0.3
 - d. 0.4
-

MCQ — Conditional probability

- 67 In a class, $P(\text{science}) = 0.65$ and $P(\text{math and science}) = 0.3$. Find $P(\text{math} \mid \text{science})$.
- a. 0.65
 - b. 0.195
 - c. ≈ 0.46
 - d. 0.3
- 68 A box has 12 balls: 7 red and 5 blue. Two balls are drawn without replacement. Given the first is blue, find $P(\text{second is blue})$.
- a. $\frac{5}{11}$
 - b. $\frac{4}{11}$
 - c. $\frac{5}{12}$
 - d. $\frac{4}{12}$
- 69 Given $P(A) = 0.6$, $P(B) = 0.4$, $P(A \text{ and } B) = 0.2$, find $P(A \mid B)$.
- a. 0.3
 - b. 0.33
 - c. 0.8
 - d. 0.5

70 Given $P(A) = 0.45$, $P(B) = 0.5$, $P(A \text{ and } B) = 0.2$, find $P(B | A)$.

- a. 0.4
- b. 0.09
- c. 0.5
- d. ≈ 0.44

MCQ — Counting: permutations and combinations

71 In how many ways can 5 books be arranged on a shelf?

- a. 25
- b. 120
- c. 20
- d. 60

72 How many ways can a committee of 4 be chosen from 9 people (order doesn't matter)?

- a. 126
- b. 3,024
- c. 362,880
- d. 36

73 A code uses 4 distinct digits from 0–9, where order matters. How many possible codes are there?

- a. 10,000
- b. 40
- c. 210
- d. 5,040

74 In how many ways can 6 people be arranged in a line?

- a. 720
- b. 120
- c. 36
- d. 6

MCQ — The five-number summary

75 For the data set 5, 9, 13, 17, 21, 25, 29, find the five-number summary (minimum, Q_1 , median, Q_3 , maximum).

- a. 9, 13, 17, 21, 25
- b. 5, 9, 17, 25, 29
- c. 5, 13, 17, 21, 29
- d. 5, 9, 13, 25, 29

76 Find the IQR for the same data set.

- a. 8
- b. 24
- c. 16
- d. 20

77 Outlier boundaries are $Q_1 - 1.5 \times IQR$ and $Q_3 + 1.5 \times IQR$. Find both boundaries for the same data set.

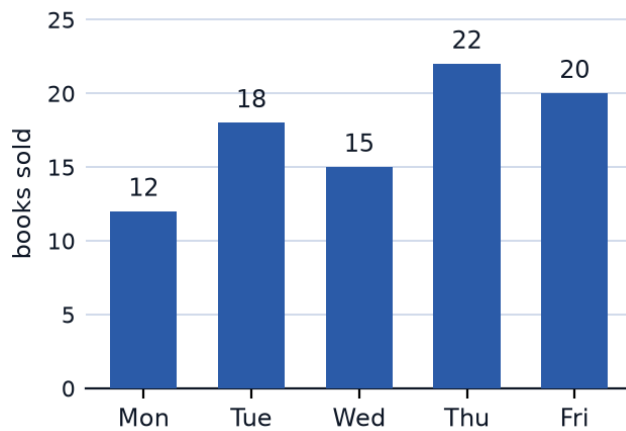
- a. -15 and 49
- b. 9 and 25
- c. -15 and 40
- d. -24 and 40

78 For the data set 10, 15, 20, 25, 30, 35, 40, find the median.

- a. 25
- b. 30
- c. 20
- d. 27.5

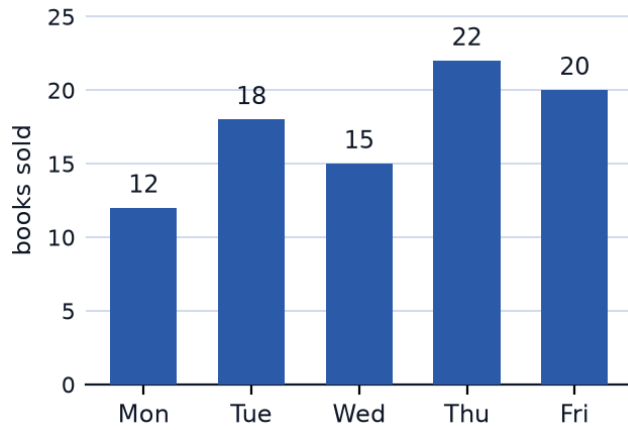
MCQ — Reading data from a bar chart

79 The bar chart shows the number of books sold each day at a bookstore over five days. Find the total books sold.



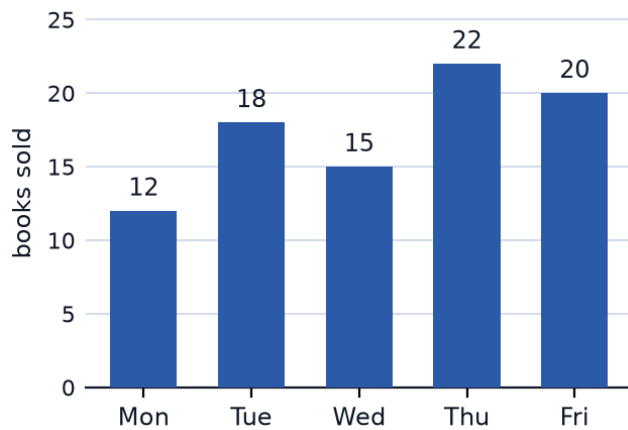
- a. 87
- b. 22
- c. 77
- d. 97

80 Using the same bar chart, find the mean number of books sold per day.



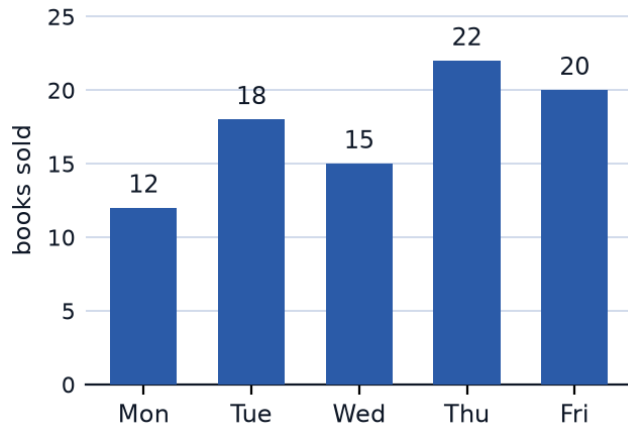
- a. 15
- b. 18
- c. 20
- d. 17.4

81 Using the same bar chart, find the range of books sold.



- a. 12
- b. 22
- c. 8
- d. 10

82 Using the same bar chart, on which day were the most books sold?



- a. Wednesday
- b. Thursday
- c. Tuesday
- d. Friday

MCQ — Word problems: statistics and probability in context

83 A survey of 250 students found that 150 like math and 100 like science, with 40 liking both. Find the probability that a randomly selected student likes math or science.

- a. 0.84
- b. 0.6
- c. 0.16
- d. 1.0

- 84 A quality-control inspector finds that 3% of items from a production line are defective. If 500 items are inspected, how many are expected to be defective?
- a. 150
 - b. 15
 - c. 3
 - d. 30
- 85 In a survey of 180 people, 108 prefer coffee over tea. Find the probability, as a percent, that a randomly selected person prefers coffee.
- a. 60%
 - b. 50%
 - c. 108%
 - d. 40%
-

Multiple choice — easy

- 86 Find the mean of 11, 1, 2, 8, 18.
- a. 7
 - b. 40
 - c. 8
 - d. 9
- 87 Find the mean of 2, 6, 1, 13, 8.
- a. 30
 - b. 6
 - c. 7
 - d. 5

- 88 Find the mean of 2, 22, 7, 12, 27.
- a. 14
 - b. 13
 - c. 15
 - d. 70
- 89 Find the mean of 17, 30, 23, 3, 7.
- a. 16
 - b. 17
 - c. 80
 - d. 15
- 90 Find the mean of 19, 25, 16, 14, 16.
- a. 90
 - b. 17
 - c. 18
 - d. 19
- 91 A bag contains 4 red and 2 blue marbles. Find $P(\text{blue})$.
- a. $\frac{1}{4}$
 - b. $\frac{1}{3}$
 - c. $\frac{1}{2}$
 - d. $\frac{2}{3}$

92 A bag contains 6 red and 3 blue marbles. Find $P(\text{blue})$.

a. $\frac{3}{11}$

b. $\frac{2}{3}$

c. $\frac{4}{9}$

d. $\frac{1}{3}$

93 Two fair coins are flipped. Find $P(\text{one head one tail})$.

a. $\frac{1}{4}$

b. $\frac{1}{2}$

c. $\frac{3}{4}$

d. $\frac{1}{8}$

94 A fair six-sided die is rolled. Find $P(\text{number} < 3)$.

a. $\frac{2}{7}$

b. $\frac{1}{3}$

c. $\frac{2}{3}$

d. $\frac{1}{2}$

95 A card is drawn from a standard 52-card deck. Find $P(\text{ace})$.

a. $\frac{3}{26}$

b. $\frac{12}{13}$

c. $\frac{1}{13}$

d. $\frac{4}{51}$

96 Find the range of 1, 12, 25, 17, 19.

a. 24

b. 22

c. 26

d. 25

97 Find the range of 28, 2, 14, 37, 32.

a. 35

b. 37

c. 38

d. 33

98 Find the range of 14, 39, 4, 32, 20.

a. 39

b. 35

c. 37

d. 33

99 Find the range of 33, 17, 7, 15, 18.

a. 26

b. 24

c. 28

d. 33

100 Find the range of 2, 1, 19, 5, 8.

- a. 20
- b. 16
- c. 19
- d. 18

Multiple choice — medium

101 Find the median of 2, 19, 28, 15, 50.

- a. 19
- b. 22
- c. 17
- d. 21

102 Find the median of 16, 40, 3, 43, 15, 26, 50.

- a. 28
- b. 27
- c. 26
- d. 24

103 Find the median of 22, 47, 37, 14, 7.

- a. 22
- b. 20
- c. 24
- d. 25

104 Find the median of 22, 11, 24, 37, 22.

- a. 23
- b. 24
- c. 20
- d. 22

105 Find the median of 22, 15, 1, 21, 39.

- a. 21
- b. 23
- c. 19
- d. 24

106 Two independent events have $P(A) = 0.7$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.

- a. 1.2
- b. 0.35
- c. 0.3
- d. 0.4

107 Two independent events have $P(A) = 0.8$ and $P(B) = 0.4$. Find $P(A \text{ and } B)$.

- a. 0.37
- b. 0.32
- c. 1.2
- d. 0.27

108 Two independent events have $P(A) = 0.8$ and $P(B) = 0.6$. Find $P(A \text{ and } B)$.

- a. 1.4
- b. 0.48
- c. 0.53
- d. 0.43

109 Two independent events have $P(A) = 0.5$ and $P(B) = 0.8$. Find $P(A \text{ and } B)$.

- a. 0.45
- b. 1.3
- c. 0.4
- d. 0.35

110 Two independent events have $P(A) = 0.7$ and $P(B) = 0.8$. Find $P(A \text{ and } B)$.

- a. 0.56
- b. 1.5
- c. 0.61
- d. 0.51

111 If $P(A) = 0.8$, find $P(\text{not } A)$.

- a. 0.8
- b. 0.3
- c. 0.2
- d. 0.1

112 If $P(A) = 0.2$, find $P(\text{not } A)$.

- a. 0.7
- b. 0.9
- c. 0.8
- d. 0.2

113 If $P(A) = 0.4$, find $P(\text{not } A)$.

- a. 0.7
- b. 0.6
- c. 0.5
- d. 0.4

114 If $P(A) = 0.3$, find $P(\text{not } A)$.

- a. 0.7
- b. 0.3
- c. 0.8
- d. 0.6

115 If $P(A) = 0.4$ and $P(B) = 0.4$, and A, B are mutually exclusive, find $P(A \text{ or } B)$.

- a. 0.8
- b. 0.16
- c. 0.0
- d. 0.9

Multiple choice — hard

116 A data set has variance 100. State its standard deviation.

- a. 9
- b. 11
- c. 100
- d. 10

117 A data set has variance 36. State its standard deviation.

- a. 36
- b. 5
- c. 6
- d. 7

118 Find the variance of 1, 3, 5, 7, 9, using $\sigma^2 = \frac{\sum(x - \mu)^2}{n}$.

- a. 16
- b. 13
- c. 8
- d. 5

119 A data set has variance 49. State its standard deviation.

- a. 8
- b. 7
- c. 6
- d. 49

- 120 Find the variance of 10, 20, 30, 40, 50, using $\sigma^2 = \frac{\sum (x - \mu)^2}{n}$.
- a. 200
 - b. 205
 - c. 400
 - d. 30
- 121 A box has 12 balls: 4 red and 8 blue. Two balls are drawn without replacement. Find $P(\text{2nd is red} \mid \text{1st is red})$.
- a. $\frac{3}{11}$
 - b. $\frac{1}{3}$
 - c. $\frac{4}{11}$
 - d. $\frac{8}{11}$
- 122 A box has 10 balls: 4 red and 6 blue. Two balls are drawn without replacement. Find $P(\text{2nd is red} \mid \text{1st is red})$.
- a. $\frac{4}{9}$
 - b. $\frac{2}{3}$
 - c. $\frac{1}{3}$
 - d. $\frac{2}{5}$
- 123 A box has 12 balls: 3 red and 9 blue. Two balls are drawn without replacement. Find $P(\text{2nd is red} \mid \text{1st is red})$.
- a. $\frac{1}{4}$
 - b. $\frac{3}{11}$
 - c. $\frac{2}{11}$
 - d. $\frac{9}{11}$

124 Given $P(A) = 0.4$ and $P(A \text{ and } B) = 0.2$, find $P(B | A)$.

- a. 0.4
- b. 0.6
- c. 0.5
- d. 2.0

125 A box has 8 balls: 5 red and 3 blue. Two balls are drawn without replacement. Find $P(\text{2nd is red} | \text{1st is red})$.

- a. $\frac{4}{7}$
- b. $\frac{5}{7}$
- c. $\frac{3}{7}$
- d. $\frac{5}{8}$

126 A code uses 3 distinct digits from 0 to 9, where order matters. How many possible codes are there?

- a. 710
- b. 730
- c. 1000
- d. 720

127 How many ways can a committee of 2 be chosen from 6 people (order doesn't matter)?

- a. 17
- b. 30
- c. 13
- d. 15

128 How many ways can a committee of 3 be chosen from 7 people (order doesn't matter)?

- a. 33
- b. 210
- c. 37
- d. 35

129 In how many ways can 6 books be arranged on a shelf?

- a. 726
- b. 720
- c. 714
- d. 36

130 How many ways can a committee of 3 be chosen from 9 people (order doesn't matter)?

- a. 82
- b. 504
- c. 86
- d. 84